From the ESKF to invariant and tangent-group filters

A handnote in matrix form, illustrated with the `kf-toy` teaching examples

Review draft 3 — figures generated from `results/teaching` (4 seeds, 12 s, `kf_teaching-v2`)

What this note covers

The note walks one estimation problem — IMU propagation with body-velocity and heading corrections — through five formulations. Each step changes a small number of matrices and leaves the rest of the filter untouched, and each chapter shows only the matrices that change, followed by the toy-example figures that make the change visible. Nothing in a chapter depends on a later one.

Why a ladder and not a jump. The end point, the equivariant filter on the tangent group, differs from a textbook ESKF in four independent ways at once: how the IMU is integrated, how the error is defined, how the biases are parametrized, and how large the state is. A single A/B comparison between the two cannot say which of the four produced a difference in the metrics. The recipes are therefore ordered so that each rung changes exactly one of them while the previous choices are kept, and the toy's equivalence tests confirm that a rung changes nothing else (for example Step3 and the reduced tangent-group filter C15A are identical to 10^{-8}). The right-invariant chapter is both a filter in its own right — it is the invariant EKF with fixed gravity, usable as an "invariant ESKF" — and the rung without which the tangent-group chapter would change three things at once.

stage	recipes	what is new	the one thing this rung isolates	what the toy shows	ch.
ESKF	Step1	nominal flow, error vector, $\boldsymbol{F}$, $\boldsymbol{Q}$, Joseph update, reset	— (baseline)	the baseline is consistent when the model matches	1
ESKF + Galilean flow	Step2	one matrix exponential replaces Euler; Γ blocks in $\boldsymbol{F}$	IMU integration formula; error definition unchanged	exact for held input; input sampling is the remaining error	2
Invariant error state	Step3, Step4	right-invariant error on the extended pose; physical and reset Jacobians	error definition (Step3), then bias parametrization (Step4); flow unchanged	propagation no longer depends on the attitude estimate	3
EqF on the tangent group	Step5A, C15A Step5B, C15B	9-vector bias with a virtual part; constraint handling; midpoint output	state extension by $\boldsymbol{b}_\nu$ (5A vs Step4); output row only (B vs A)	constant coupling; how $\boldsymbol{b}_\nu$ is kept at zero; what one output row changes	4, 5
Iterated solver	GN, LM for every recipe	fixed-prior MAP; the KF as the first Gauss–Newton step	number of linearizations; prior, likelihood and chart unchanged	large-error recovery; output variants coincide under iteration	6
Closing	—	diagnostics under model mismatch	the model, not the geometry	NEES and coverage expose what cost and NIS do not	7

Notation, the operators $[\cdot]_\times$, Γ_1, Γ_2, Ad, ad, the physical error chart used for every score, and the common experiment are collected in Appendix A: and linked at first use. Two conventions hold throughout: t is time and h is the length of one propagation interval; a hat marks an estimate, no hat the true value.

Contents

What this note covers 1
1 The error-state Kalman filter (Step1) 3
1.1 Nominal propagation with a held input 3
1.2 Error definition: body-frame attitude, additive everything else 3
1.3 Discrete transition and process noise 3
1.4 Measurement update and the reset 4
1.5 Measurement Jacobians in this chart 4
1.6 Step1 in the toy 4
2 ESKF with the Galilean flow (Step2) 5
2.1 The navigation state as a matrix 5
2.2 Held IMU input is a single matrix exponential 5
2.3 The transition matrix gains bias-sensitivity blocks 6
2.4 What exactness does and does not buy 6
3 Invariant error state (Step3, Step4) 7
3.1 Why this step, and why now 7
3.2 Retract and local 7
3.3 The physical Jacobian 8
3.4 The reset Jacobian 8
3.5 The discrete transition in the new chart 9
3.6 The right-invariant error and its dynamics 9
3.7 Step4: bias coordinates that move with the state 9
3.8 Step1–Step4 in the toy 10
4 EqF on the tangent group (Step5A, C15A) 11
4.1 Why extend the state 11
4.2 TG error dynamics: the constant coupling 11
4.3 TG retraction and physical Jacobian 11
4.4 The constraint 11
4.5 C15: the reduced embedding 12
4.6 Step3, Step4, Step5A, C15A in the toy 13
5 Output approximation: the midpoint (C-star) output (Step5B, C15B) 14
5.1 The body-velocity measurement in the invariant chart 14
5.2 Remainder orders 14
5.3 A versus B in the toy (KF) 15
6 Iterated solver: GN and LM as fixed-prior MAP 15
6.1 The KF update is one Gauss–Newton step 15
6.2 Posterior covariance 16
6.3 Why the prior must stay fixed 16
6.4 GN/LM across all recipes in the toy 16
7 Reading the diagnostics: when the model is wrong 17
7.1 Cost is not truth 17
7.2 An unmodelled bias step 18
7.3 Full 3-D velocity versus forward + NHC 18
Appendix A: Common ground: state, operators and the yardstick 20
A.1 State, frames and IMU model 20
A.2 Operators 20
A.3 The physical chart and the scores 20
A.4 The common experiment 21
Appendix B: Symbol to code 21
Appendix C: Regenerating the figures 22
Appendix D: Mean metrics, 4 seeds, 12 s 22

1 The error-state Kalman filter (Step1)

The state is $(R, v, p, \boldsymbol{b}_g, \boldsymbol{b}_a)$ with the IMU model of A.1; the skew matrix $[\cdot]_\times$ and the series Γ_1, Γ_2 are defined in A.2. This chapter writes down the five matrices that every later chapter will modify.

1.1 Nominal propagation with a held input

Over one interval of length h with the held, bias-corrected input $w = \tilde{w} - \hat{\boldsymbol{b}}_g$, $a = \tilde{a} - \widehat{\boldsymbol{b}_a}$, the Euler ESKF propagates

$$\begin{bmatrix} R^+ \\ v^+ \\ p^+ \\ \boldsymbol{b}_g^+ \\ \boldsymbol{b}_a^+ \end{bmatrix} = \begin{bmatrix} R\exp(h[w]_\times) \\ v + hg + hRa \\ p + hv + \frac{1}{2}h^2 g + \frac{1}{2}h^2 Ra \\ \boldsymbol{b}_g \\ \boldsymbol{b}_a \end{bmatrix}. \tag{1}$$

1.2 Error definition: body-frame attitude, additive everything else

The true state is the estimate perturbed by the 15-vector $\delta\boldsymbol{x} = (\delta\boldsymbol{\theta}, \delta v, \delta p, \delta\boldsymbol{b}_g, \delta\boldsymbol{b}_a)$:

$$\begin{bmatrix} R \\ v \\ p \\ \boldsymbol{b}_g \\ \boldsymbol{b}_a \end{bmatrix} = \begin{bmatrix} \hat{\boldsymbol{R}}\exp([\delta\boldsymbol{\theta}]_\times) \\ \hat{v} + \delta v \\ \hat{p} + \delta p \\ \hat{\boldsymbol{b}}_g + \delta\boldsymbol{b}_g \\ \widehat{\boldsymbol{b}_a} + \delta\boldsymbol{b}_a \end{bmatrix}. \tag{2}$$

This is the **left** (body-frame) perturbation. It coincides with the physical error vector used for all scores (A.3), so for Step1 no coordinate change is needed before scoring.

1.3 Discrete transition and process noise

Linearizing Eq. 1 in the coordinates of Eq. 2:

$$\boldsymbol{F}_{\text{Step1}} = \begin{bmatrix} \exp(-h[w]_\times) & 0 & 0 & -h\Gamma_1(-hw) & 0 \\ -h\hat{\boldsymbol{R}}[a]_\times & \boldsymbol{I}_3 & 0 & 0 & -h\hat{\boldsymbol{R}} \\ -\frac{1}{2}h^2\hat{\boldsymbol{R}}[a]_\times & h\boldsymbol{I}_3 & \boldsymbol{I}_3 & 0 & -\frac{1}{2}h^2\hat{\boldsymbol{R}} \\ 0 & 0 & 0 & \boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{I}_3 \end{bmatrix}. \tag{3}$$

Continuous white noise $\boldsymbol{n} = (\boldsymbol{n}_g, \boldsymbol{n}_a, \boldsymbol{n}_{bg}, \boldsymbol{n}_{ba})$ enters through

$$\boldsymbol{G} = \begin{bmatrix} -\boldsymbol{I}_3 & 0 & 0 & 0 \\ 0 & -\hat{\boldsymbol{R}} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & \boldsymbol{I}_3 \end{bmatrix}, \qquad \boldsymbol{Q}_c = \begin{bmatrix} \sigma_g^2\boldsymbol{I}_3 & 0 & 0 & 0 \\ 0 & \sigma_a^2\boldsymbol{I}_3 & 0 & 0 \\ 0 & 0 & \sigma_{bg}^2\boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & \sigma_{ba}^2\boldsymbol{I}_3 \end{bmatrix}, \tag{4}$$

and the discrete process covariance transports the noise injected at time $t \in [0, h]$ to the end of the interval:

$$\boldsymbol{Q} = \int_0^h \boldsymbol{\Phi}(h, t)\boldsymbol{G}(t)\boldsymbol{Q}_c\boldsymbol{G}(t)^\top\boldsymbol{\Phi}(h, t)^\top \, \mathrm{d}t. \tag{5}$$

The toy evaluates Eq. 5 with three-point Gauss quadrature; the tests check it against integrated white acceleration,

$$\begin{bmatrix} \boldsymbol{Q}_{vv} & \boldsymbol{Q}_{vp} \\ \boldsymbol{Q}_{pv} & \boldsymbol{Q}_{pp} \end{bmatrix} = \sigma_a^2 \begin{bmatrix} h\boldsymbol{I}_3 & \frac{1}{2}h^2\boldsymbol{I}_3 \\ \frac{1}{2}h^2\boldsymbol{I}_3 & \frac{1}{3}h^3\boldsymbol{I}_3 \end{bmatrix}. \tag{6}$$

The engine never forms $\boldsymbol{F}\boldsymbol{P}\boldsymbol{F}^\top + \boldsymbol{Q}$ explicitly. It carries a square root $\boldsymbol{L}$, $\boldsymbol{P} = \boldsymbol{L}\boldsymbol{L}^\top$, and compresses

$$[\boldsymbol{F}\boldsymbol{L} \;\; \boldsymbol{N}] \xrightarrow{\text{QR}} \boldsymbol{L}^+, \qquad \boldsymbol{N}\boldsymbol{N}^\top = \boldsymbol{Q}, \tag{7}$$

so that positive semidefiniteness is structural.

1.4 Measurement update and the reset

For a residual $\boldsymbol{r} = \boldsymbol{z} - \boldsymbol{y}(\hat{\boldsymbol{x}})$ with predicted measurement $\boldsymbol{y}$, Jacobian $\boldsymbol{H}$ and noise $\boldsymbol{V}$:

$$\begin{bmatrix} \boldsymbol{S} \\ \boldsymbol{K} \\ \delta\boldsymbol{x} \\ \boldsymbol{P}' \end{bmatrix} = \begin{bmatrix} \boldsymbol{H}\boldsymbol{P}\boldsymbol{H}^\top + \boldsymbol{V} \\ \boldsymbol{P}\boldsymbol{H}^\top\boldsymbol{S}^{-1} \\ \boldsymbol{K}\boldsymbol{r} \\ (I - \boldsymbol{K}\boldsymbol{H})\boldsymbol{P}(I - \boldsymbol{K}\boldsymbol{H})^\top + \boldsymbol{K}\boldsymbol{V}\boldsymbol{K}^\top \end{bmatrix}. \tag{8}$$

The correction is retracted into the nominal state with Eq. 2. Because the attitude error was defined around the **old** estimate, the covariance is transported to the new one:

$$\boldsymbol{P}^+ = \boldsymbol{J}\boldsymbol{P}'\boldsymbol{J}^\top, \qquad \boldsymbol{J}_{\text{exact}} = \begin{bmatrix} \boldsymbol{J}_r(\delta\boldsymbol{\theta}) & 0 \\ 0 & \boldsymbol{I}_{12} \end{bmatrix}, \quad \boldsymbol{J}_{\text{1st}} = \begin{bmatrix} \boldsymbol{I}_3 - \frac{1}{2}[\delta\boldsymbol{\theta}]_\times & 0 \\ 0 & \boldsymbol{I}_{12} \end{bmatrix}, \quad \boldsymbol{J}_{\text{textbook}} = \boldsymbol{I}_{15}, \tag{9}$$

with $\boldsymbol{J}_r(\delta\boldsymbol{\theta}) = \Gamma_1(-\delta\boldsymbol{\theta})$ the right Jacobian of SO(3). All recipes in the toy use the exact form; the size of the first-order error is plotted in Section 2.

1.5 Measurement Jacobians in this chart

With $\hat{v}_B = \hat{\boldsymbol{R}}^\top\hat{v}$ the predicted body velocity, lever arm $\boldsymbol{l}$, and $\boldsymbol{h}_\theta$ the derivative of the yaw of the mounted x-axis with respect to $\delta\boldsymbol{\theta}$:

$$\begin{bmatrix} \boldsymbol{H}_{\text{vel}} \\ \boldsymbol{H}_{\text{pos}} \\ \boldsymbol{H}_{\text{yaw}} \end{bmatrix} = \begin{bmatrix} [\hat{v}_B]_\times & \hat{\boldsymbol{R}}^\top & 0 & 0 & 0 \\ -\hat{\boldsymbol{R}}[\boldsymbol{l}]_\times & 0 & \boldsymbol{I}_3 & 0 & 0 \\ \boldsymbol{h}_\theta^\top & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{10}$$

In every scenario of this note $\boldsymbol{H}_{\text{vel}}$ receives a genuine three-axis body-velocity measurement with noise on all three channels. A wheel-speed or CAN sensor supplies only the forward channel; what is usually done about the other two is the non-holonomic constraint, discussed on its own in Section 7.3.

1.6 Step1 in the toy

The baseline scenario is `matched_heading` (A.4): fixed biases, matched noise, heading every 0.2 s.

How to read NEES, NIS and coverage
A filter reports an error covariance $\boldsymbol{P}$ as well as an estimate; the question is whether $\boldsymbol{P}$ is honest. The **normalized estimation error squared** compares the true error (available in the toy) with that covariance,

$$\text{NEES}_{15} = \boldsymbol{e}^\top\boldsymbol{P}_{\text{phys}}^{-1}\boldsymbol{e}, \tag{11}$$

with $\boldsymbol{e}$ the 15-dimensional physical error of A.3. If the error is Gaussian with covariance $\boldsymbol{P}$, NEES_{15} is χ^2 with 15 degrees of freedom and has mean 15; the toy reports its time average divided by 15, so a **consistent** filter sits near 1, a value above 1 means the filter is **overconfident** (its $\boldsymbol{P}$ is too small for its actual error) and a value below 1 means it is **conservative**. NEES says nothing about how small the error is — a filter can have low RMSE and NEES of 70, or high RMSE and NEES of 1. The **normalized innovation squared**

is the same construction applied to the measurement residual, NIS $= \boldsymbol{r}^\top \boldsymbol{S}^{-1}\boldsymbol{r}$ with $\boldsymbol{S}$ from Eq. 8; it needs no truth and has mean 3 for a 3-D velocity, but it only sees what the measurement sees. **Coverage** is the fraction of time the true error of one component lies inside its $\pm 1.96\sigma$ band; it should be 0.95. Time samples are correlated, so these are diagnostics, not significance tests.



Figure 1: Step1 KF position error, seed 0. Errors stay at the 0.1–0.3 m level set by the velocity noise and the prior.

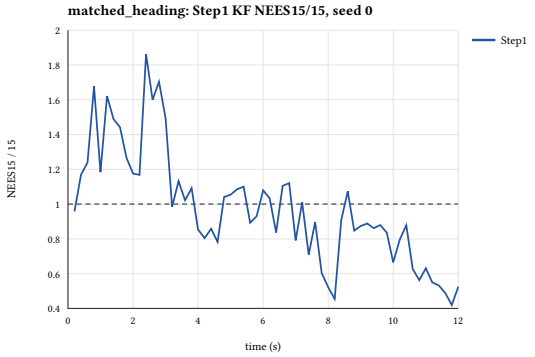


Figure 2: Step1 KF NEES15/15 over time, seed 0. It fluctuates around 1; the four-seed mean is 0.954.

Mean over 4 seeds, Step1 KF: position RMSE 0.179 m, attitude RMSE 0.0162 rad, NEES15/15 0.954, velocity NIS 2.90 (3 d.o.f.).

In one line
The ESKF is five matrices: flow Eq. 1, chart Eq. 2, transition Eq. 3 with noise Eq. 4–Eq. 5, update Eq. 8, reset Eq. 9. When noise and model match the data it is consistent (NEES ≈ 1, NIS ≈ 3).

2 ESKF with the Galilean flow (Step2)

Step2 keeps the chart Eq. 2, the update Eq. 8 and the reset Eq. 9. It replaces the flow Eq. 1 and therefore the transition Eq. 3. Writing the navigation state as one matrix shows why the new flow is exact.

2.1 The navigation state as a matrix

$$\boldsymbol{X} = \begin{bmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in \mathrm{SE}_2(3) \subset \mathbb{R}^{5\times 5}, \qquad \boldsymbol{X}_1\boldsymbol{X}_2 = \begin{bmatrix} R_1R_2 & R_1v_2 + v_1 & R_1p_2 + p_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{12}$$

The product composes rotations and transports the two vectors. The Galilean group $\mathrm{SGal}(3)$ adds a time entry in position $(2,3)$; that entry is what couples velocity into position inside a single exponential.

2.2 Held IMU input is a single matrix exponential

Put the held body input into a matrix with a unit time rate:

$$\boldsymbol{W} = \begin{bmatrix} [w]_\times & a & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \exp(h\boldsymbol{W}) = \begin{bmatrix} \exp(h[w]_\times) & h\Gamma_1(hw)a & h^2\Gamma_2(hw)a \\ 0 & 1 & h \\ 0 & 0 & 1 \end{bmatrix}. \tag{13}$$

The closed form follows from $\boldsymbol{W}^k$ having $[w]_\times^{k-1}a$ in the velocity column and $[w]_\times^{k-2}a$ in the position column; summing with $1/k!$ gives exactly the Γ series of A.2. With gravity as a left factor on a state whose time entry is zero at the start of the interval,

$$\boldsymbol{X}^{+} = \underbrace{\begin{bmatrix} \boldsymbol{I}_3 & h\boldsymbol{g} & -\frac{1}{2}h^2\boldsymbol{g} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{gravity}} \boldsymbol{X} \underbrace{\exp(h\boldsymbol{W})}_{\text{held body input}}, \tag{14}$$

which multiplies out to the Step2 flow

$$\begin{bmatrix} R^{+} \\ v^{+} \\ p^{+} \\ \boldsymbol{b}_g^{+} \\ \boldsymbol{b}_a^{+} \end{bmatrix} = \begin{bmatrix} R\exp(h[w]_\times) \\ v + h\boldsymbol{g} + hR\Gamma_1(hw)a \\ p + hv + \frac{1}{2}h^2\boldsymbol{g} + h^2R\Gamma_2(hw)a \\ \boldsymbol{b}_g \\ \boldsymbol{b}_a \end{bmatrix}. \tag{15}$$

Against Eq. 1: Euler keeps the $k = 0$ term of each series, $\Gamma_1 \approx \boldsymbol{I}_3$, $\Gamma_2 \approx \frac{1}{2}\boldsymbol{I}_3$. For a constant body input over the interval Eq. 15 is exact for any h; the math test checks it at $h = 0.5$ s, $\omega_z = 2$ rad/s, $f_x = 3\text{m/s}^2$.

2.3 The transition matrix gains bias-sensitivity blocks

Differentiating Eq. 15 in the same chart:

$$\boldsymbol{F}_{\text{Step2}} = \begin{bmatrix} \exp(-h[w]_\times) & 0 & 0 & -h\Gamma_1(-hw) & 0 \\ -\hat{\boldsymbol{R}}[h\Gamma_1 a]_\times & \boldsymbol{I}_3 & 0 & -h^2\hat{\boldsymbol{R}}\boldsymbol{D}_1 & -h\hat{\boldsymbol{R}}\Gamma_1 \\ -\hat{\boldsymbol{R}}[h^2\Gamma_2 a]_\times & h\boldsymbol{I}_3 & \boldsymbol{I}_3 & -h^3\hat{\boldsymbol{R}}\boldsymbol{D}_2 & -h^2\hat{\boldsymbol{R}}\Gamma_2 \\ 0 & 0 & 0 & \boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{I}_3 \end{bmatrix}, \qquad \boldsymbol{D}_i = \frac{\partial(\Gamma_i(\varphi)a)}{\partial\varphi}\Big|_{\varphi=hw}. \tag{16}$$

Two blocks changed relative to Eq. 3: the accelerometer-bias column is rotated by Γ_1, Γ_2 instead of $\boldsymbol{I}_3, \frac{1}{2}\boldsymbol{I}_3$, and the gyro-bias column now reaches velocity and position through $\boldsymbol{D}_1, \boldsymbol{D}_2$ (a gyro bias tilts the integrated acceleration). $\boldsymbol{G}$ and Eq. 5 are unchanged.

2.4 What exactness does and does not buy

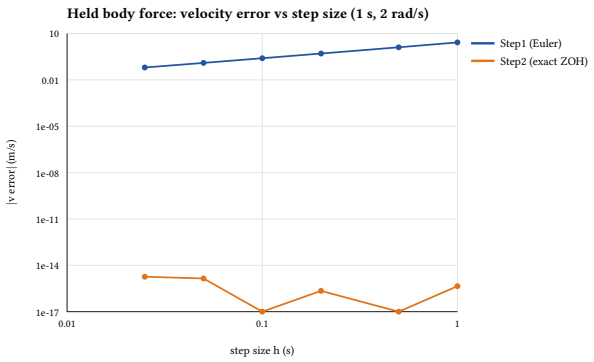


Figure 3: Constant body force rotating at 2 rad/s, integrated over 1 s. Step2 is exact at every step size (the floor is round-off); Step1 converges at first order.



Figure 4: Body force $3\cos 1.7t$. Both recipes now carry input-sampling error. Holding the **midpoint** sample lets Step2 converge at second order; the left sample does not.

Once the real signal varies between samples, the integration formula stops being the bottleneck: the description of the input over the interval is. Exact ZOH applied to the wrong sample is still first order.

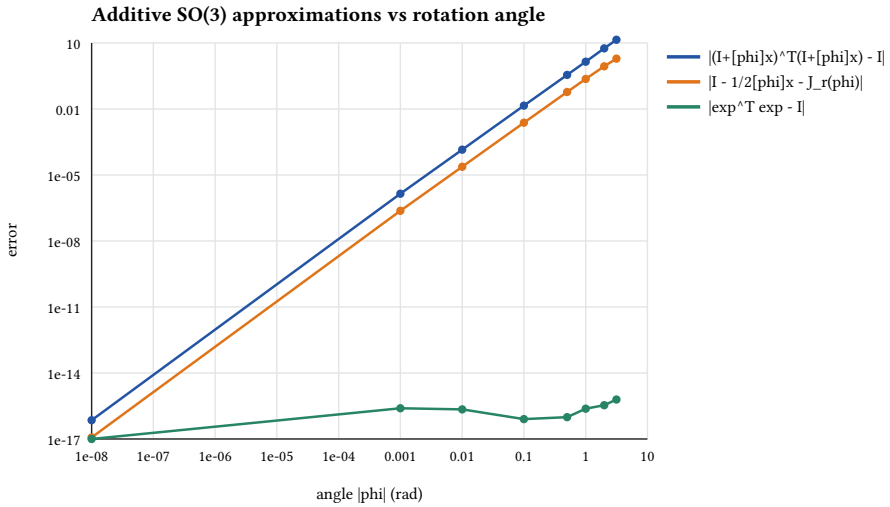


Figure 5: Additive rotation approximations vs. angle. $I + [\boldsymbol{\varphi}]_\times$ loses orthogonality quadratically; the first-order reset $I - \frac{1}{2}[\boldsymbol{\varphi}]_\times$ departs from $\boldsymbol{J}_r$ cubically and is visible from 0.1 rad; exp stays orthogonal to machine precision.

On the toy scenarios at 40 Hz the two recipes are indistinguishable (KF, 4-seed mean): `matched_heading` 0.179 vs 0.181 m, `pure_dr` 0.962 vs 0.963 m, NEES15/15 0.954 vs 0.953. The step-size sweep is where the difference lives.

In one line
Step2 = Step1 with the flow replaced by the single exponential Eq. 13–Eq. 14 and the transition by Eq. 16. The error definition is untouched.

3 Invariant error state (Step3, Step4)

3.1 Why this step, and why now

Section 2 changed how the IMU is integrated and left the error definition Eq. 2 alone. This chapter does the opposite: it keeps the Step2 flow and changes only what the 15-vector $\delta\boldsymbol{x}$ **means**. Two reasons justify the rung on its own and one justifies its position.

As a filter, the result is the right-invariant EKF with fixed gravity — what is often called the invariant ESKF. Its selling point is visible directly in the matrices: the attitude and velocity rows of the error dynamics stop depending on the attitude estimate (Eq. 24 below), so a wrong attitude does not corrupt its own propagation. That property is what the `pure_dr` scenario is designed to probe.

As a rung, Step3 is the tangent-group filter of Section 4 with the virtual bias removed; the toy proves this by showing Step3 and C15A identical to 10^{-8}. Had the note gone from Step2 straight to Step5A, the comparison would have changed the error definition, the bias parametrization and the state dimension together, and no metric could have been attributed to any one of them. Step3 isolates the error definition; Step4 then isolates the bias parametrization; Section 4 isolates the state extension.

Changing what $\delta\boldsymbol{x}$ means forces three bookkeeping questions that the ESKF never had to ask, because in the left chart their answers are trivial (identity or nearly so). They are answered first, in Section 3.2–Section 3.4, so that the invariant error in Section 3.6 can be used immediately.

3.2 Retract and local

Why it is needed. The Kalman update Eq. 8 produces a **vector** $\delta\boldsymbol{x}$; the estimator state is a rotation matrix and four vectors. Something must turn the vector into a state (to apply the correction) and two states into a vector (to compute the error $\boldsymbol{e}$ for scoring, to run GN/LM later, and to check Jacobians by finite differences). In the ESKF

this was Eq. 2 and it was never named. Once the vector is allowed to mean something else, the two maps must be written down explicitly, and every other matrix in the filter is a derivative of them.

A chart is a pair of inverse maps between a state and a vector, $\boldsymbol{x} = \mathrm{retract}(\hat{\boldsymbol{x}}, \delta)$ and $\delta = \mathrm{local}(\hat{\boldsymbol{x}}, \boldsymbol{x})$. Eq. 2 is the left chart. The **right-invariant** chart multiplies the 5×5 matrix Eq. 12 from the left:

$$\boldsymbol{X} = \exp(\boldsymbol{\xi}^\wedge)\hat{\boldsymbol{X}}, \qquad \boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\xi}_\theta]_\times & \boldsymbol{\xi}_v & \boldsymbol{\xi}_p \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \exp(\boldsymbol{\xi}^\wedge) = \begin{bmatrix} \boldsymbol{E} & \Gamma_1(\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_v & \Gamma_1(\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{17}$$

with $\boldsymbol{E} = \exp\big([\boldsymbol{\xi}_\theta]_\times\big)$, so that

$$\begin{bmatrix} R \\ v \\ p \\ \boldsymbol{b}_g \\ \boldsymbol{b}_a \end{bmatrix} = \begin{bmatrix} \boldsymbol{E}\hat{\boldsymbol{R}} \\ \boldsymbol{E}\hat{v} + \Gamma_1(\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_v \\ \boldsymbol{E}\hat{p} + \Gamma_1(\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_p \\ \hat{\boldsymbol{b}}_g + \delta\boldsymbol{b}_g \\ \hat{\boldsymbol{b}}_a + \delta\boldsymbol{b}_a \end{bmatrix}. \tag{18}$$

Biases stay additive in Step3; Step4 changes them too (Eq. 26).

3.3 The physical Jacobian

Why it is needed. After the chart changes, the entry P_{44} of Step3′s covariance is no longer "variance of the x-velocity error" in the sense of A.3: the native coordinates $\boldsymbol{\xi}_v$ mix velocity with a world-frame rotation of the estimate. Comparing Step3 against Step1, computing NEES against the same truth, or plotting a $\pm 1.96\sigma$ band for the z position, all require moving the covariance into one agreed chart first. Without this matrix the recipes would be scored in different units and any ranking between them would be meaningless.

The native covariance $\boldsymbol{P}$ now lives in $\boldsymbol{\xi}$ coordinates while every score lives in the physical chart (A.3). To first order $\delta\boldsymbol{x} = \boldsymbol{J}_{\text{phys}}\boldsymbol{\xi}$ with

$$\boldsymbol{J}_{\text{phys}} = \begin{bmatrix} \hat{\boldsymbol{R}}^\top & 0 & 0 & 0 & 0 \\ -[\hat{v}]_\times & \boldsymbol{I}_3 & 0 & 0 & 0 \\ -[\hat{p}]_\times & 0 & \boldsymbol{I}_3 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{I}_3 \end{bmatrix}, \qquad \boldsymbol{P}_{\text{phys}} = \boldsymbol{J}_{\text{phys}}\boldsymbol{P}\boldsymbol{J}_{\text{phys}}^\top. \tag{19}$$

The $\hat{\boldsymbol{R}}^\top$ block says a world-frame rotation perturbation is a body-frame one rotated back; the $-[\hat{v}]_\times$, $-[\hat{p}]_\times$ blocks say that a world-frame rotation also moves velocity and position. A diagonal native $\boldsymbol{P}$ in this chart is a **correlated** physical covariance. The math test checks Eq. 19 against central finite differences for all recipes.

3.4 The reset Jacobian

Why it is needed. The covariance before an update describes a perturbation **around the old estimate**. The update moves the estimate by a finite δ; the tangent space moves with it, and a perturbation of the old estimate is not the same vector as the equally sized perturbation of the new one. For the left chart the mismatch is only in the attitude block and is the $\boldsymbol{J}_r(\delta\boldsymbol{\theta})$ of Eq. 9, which textbooks often drop; the SO(3) figure in Section 2 shows the dropped term growing cubically with the correction. In the right-invariant chart the mismatch couples attitude, velocity and position, so it can no longer be ignored, and the same matrix will be applied at every iterate of GN/LM in Section 6.

After a correction δ the covariance is re-expressed around the new estimate. For the two charts:

$$\boldsymbol{J}_{\text{reset}}^{\text{left}} = \begin{bmatrix} \Gamma_1(-\delta\boldsymbol{\theta}) & 0 \\ 0 & \boldsymbol{I}_{12} \end{bmatrix}, \qquad \boldsymbol{J}_{\text{reset}}^{\text{right}} = \begin{bmatrix} \sum_{k\geq 0} \mathrm{ad}\,\frac{(\boldsymbol{\xi}_{\text{nav}})^k}{(k+1)!} & 0 \\ 0 & \boldsymbol{I}_6 \end{bmatrix}, \tag{20}$$

with ad from A.2. Both are finite maps, not first-order approximations.

3.5 The discrete transition in the new chart

The toy does not derive a second transition matrix. It conjugates the physical-chart result:

$$\boldsymbol{F}_{\text{native}} = \boldsymbol{J}^{-1}_{\text{phys}(\hat{\boldsymbol{x}}^+)}\ \boldsymbol{F}_{\text{phys}}\ \boldsymbol{J}_{\text{phys}(\hat{\boldsymbol{x}})}, \qquad \boldsymbol{Q}_{\text{native}} = \boldsymbol{J}^{-1}_{\text{phys}(\hat{\boldsymbol{x}}^+)}\ \boldsymbol{Q}_{\text{phys}}\ \boldsymbol{J}^{-\top}_{\text{phys}(\hat{\boldsymbol{x}}^+)}. \tag{21}$$

Step3 therefore propagates the same physical covariance as Step2 up to linearization; a difference in results is a difference in error definition, not in integration.

3.6 The right-invariant error and its dynamics

Define the error as the group difference between true and estimated extended poses, true on the left:

$$\boldsymbol{\eta} = \boldsymbol{X}\hat{\boldsymbol{X}}^{-1} = \begin{bmatrix} R\hat{\boldsymbol{R}}^\top & v - R\hat{\boldsymbol{R}}^\top\hat{v} & p - R\hat{\boldsymbol{R}}^\top\hat{p} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{\xi} = \operatorname{Log}(\boldsymbol{\eta}). \tag{22}$$

To first order $\boldsymbol{\xi}_\theta$ is a **world-frame** attitude error and $\boldsymbol{\xi}_v \approx v - \boldsymbol{E}\hat{v}$, $\boldsymbol{\xi}_p \approx p - \boldsymbol{E}\hat{p}$ — the chart Eq. 17. Differentiating along the IMU flow with bias errors $\delta\boldsymbol{b}_g = \boldsymbol{b}_g - \hat{\boldsymbol{b}}_g$, $\delta\boldsymbol{b}_a = \boldsymbol{b}_a - \hat{\boldsymbol{b}}_a$ gives $\dot{\boldsymbol{\xi}} = \boldsymbol{A}\boldsymbol{\xi} + \boldsymbol{G}\boldsymbol{n}$ with, side by side,

$$\boldsymbol{A}_{\text{left (Step1/2)}} = \begin{bmatrix} -[w]_\times & 0 & 0 & -\boldsymbol{I}_3 & 0 \\ -\hat{\boldsymbol{R}}[a]_\times & 0 & 0 & 0 & -\hat{\boldsymbol{R}} \\ 0 & \boldsymbol{I}_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \tag{23}$$

$$\boldsymbol{A}_{\text{right-invariant (Step3)}} = \begin{bmatrix} 0 & 0 & 0 & -\hat{\boldsymbol{R}} & 0 \\ [g]_\times & 0 & 0 & -[\hat{v}]_\times\hat{\boldsymbol{R}} & -\hat{\boldsymbol{R}} \\ 0 & \boldsymbol{I}_3 & 0 & -[\hat{p}]_\times\hat{\boldsymbol{R}} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{24}$$

Read the two 9×9 navigation blocks (upper-left). In the left chart they contain the inputs w, a and the estimate $\hat{\boldsymbol{R}}$. In the right-invariant chart they contain only $[g]_\times$ and $\boldsymbol{I}_3$. The estimate has moved into the bias columns and into the noise matrix,

$$\boldsymbol{G}_{\text{Step3}} = \begin{bmatrix} -\hat{\boldsymbol{R}} & 0 & 0 & 0 \\ -[\hat{v}]_\times\hat{\boldsymbol{R}} & -\hat{\boldsymbol{R}} & 0 & 0 \\ -[\hat{p}]_\times\hat{\boldsymbol{R}} & 0 & 0 & 0 \\ 0 & 0 & \boldsymbol{I}_3 & 0 \\ 0 & 0 & 0 & \boldsymbol{I}_3 \end{bmatrix} = \begin{bmatrix} -\operatorname{Ad}_{\hat{\boldsymbol{X}}}\begin{bmatrix}\boldsymbol{I}_6\\0\end{bmatrix} & 0 \\ 0 & \boldsymbol{I}_6 \end{bmatrix}, \tag{25}$$

with Ad from A.2. Two consequences follow and are what the figures below test: the attitude row no longer depends on the current attitude estimate, so a large attitude error does not corrupt its own propagation; and world-frame measurements get simple Jacobians (used in Section 5).

3.7 Step4: bias coordinates that move with the state

Step3 leaves the biases additive, so their coupling columns in Eq. 24 still carry $\hat{\boldsymbol{R}}$, $\hat{v}$, $\hat{p}$. Step4 changes the **bias** chart:

$$\begin{bmatrix}\delta\boldsymbol{b}_g \\ \delta\boldsymbol{b}_a\end{bmatrix} = \operatorname{Ad}^{-1}_{\hat{\boldsymbol{R}},\hat{v}}\ \boldsymbol{J}_{-\operatorname{ad}}(\boldsymbol{\xi}_{\theta v})\ \begin{bmatrix}\boldsymbol{\xi}_{bg} \\ \boldsymbol{\xi}_{ba}\end{bmatrix}, \qquad \operatorname{Ad}_{\hat{\boldsymbol{R}},\hat{v}} = \begin{bmatrix}\hat{\boldsymbol{R}} & 0 \\ [\hat{v}]_\times\hat{\boldsymbol{R}} & \hat{\boldsymbol{R}}\end{bmatrix}, \qquad \boldsymbol{J}_{-\operatorname{ad}}(\boldsymbol{\xi}) = \sum_{k\geq 0}\frac{(-\operatorname{ad}(\boldsymbol{\xi}))^k}{(k+1)!}, \tag{26}$$

with the 6×6 restriction of Ad, ad. Its physical Jacobian carries $\mathrm{Ad}^{-1}_{\widehat{\boldsymbol{R}},\hat{v}}$ in the bias block and its reset Jacobian is the 15×15 series of $\begin{bmatrix} \mathrm{ad}(\boldsymbol{\xi}_{\mathrm{nav}}) & 0 \\ \mathrm{ad}(\boldsymbol{\xi}_b) & \mathrm{ad}(\boldsymbol{\xi}_{\theta v}) \end{bmatrix}$. The bias uncertainty is expressed in the same moving frame as the navigation error.

3.8 Step1–Step4 in the toy

The scenario that separates the charts is `pure_dr` (IMU and body velocity only; global yaw unobservable), with `heading_dropout` as the control.

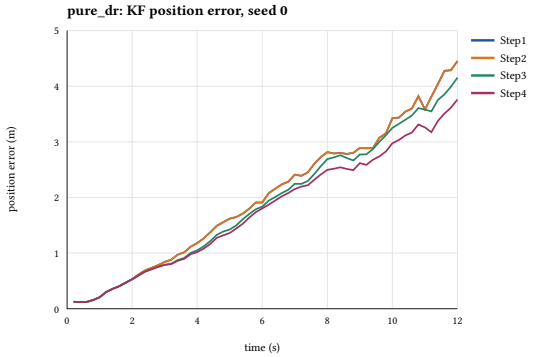


Figure 6: `pure_dr`, KF, seed 0. Position error grows for all four recipes; Step3 and Step4 sit below Step1/2 on this seed.

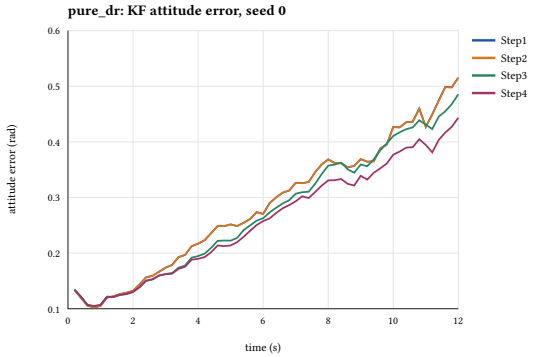


Figure 7: `pure_dr`, KF attitude error, seed 0. The unobservable yaw drifts; the invariant charts drift less on this seed.

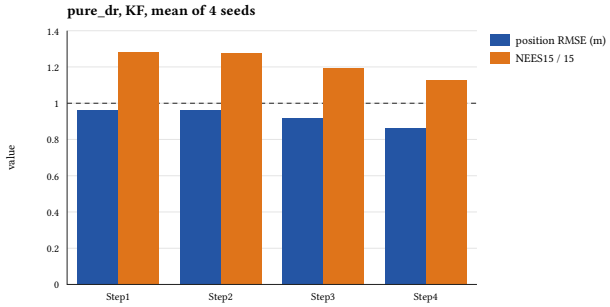


Figure 8: `pure_dr`, KF, 4-seed means. Position RMSE 0.96 / 0.96 / 0.92 / 0.87 m; NEES15/15 1.28 / 1.28 / 1.20 / 1.13 for Step1–4. All are overconfident; the invariant charts less so.

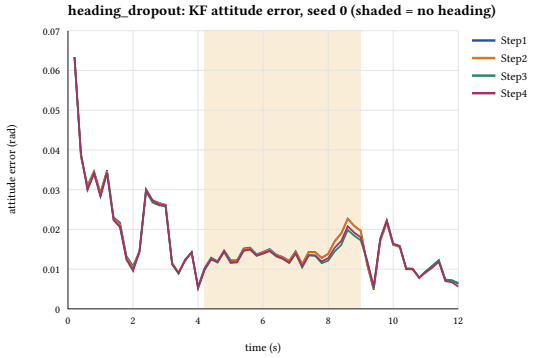


Figure 9: `heading_dropout`, KF attitude error. Inside the shaded window all four drift alike and all recover at once when heading returns: no chart supplies missing information.

Reading these figures

The `pure_dr` differences are real but small and four seeds do not establish a ranking. What the figures support: the right-invariant error improves **consistency** when attitude is poorly observed, because its propagation Eq. 24 does not depend on the attitude estimate. It does not improve **observability**.

In one line

Invariant error = move the estimate out of the navigation block of $\boldsymbol{A}$ and into the bias and noise columns (Eq. 24–Eq. 25). Reading and transporting the covariance needs Eq. 19 and Eq. 20. Step4 additionally re-expresses the biases (Eq. 26).

4 EqF on the tangent group (Step5A, C15A)

4.1 Why extend the state

In Eq. 24 the bias columns $-\hat{\boldsymbol{R}}$, $-[\hat{v}]_\times \hat{\boldsymbol{R}}$, $-[\hat{p}]_\times \hat{\boldsymbol{R}}$ are the first two column blocks of $-\mathrm{Ad}_{\widehat{\boldsymbol{X}}}$. If the bias were a full 9-vector $\boldsymbol{b} = (\boldsymbol{b}_g, \boldsymbol{b}_a, \boldsymbol{b}_\nu)$ paired with all three rows of the navigation algebra, the coupling would be $-\mathrm{Ad}$ in full, and expressing the bias error in the frame moved by Ad would make it a constant $-\boldsymbol{I}_9$. That is the tangent group $\mathrm{TG} = \mathrm{SE}_2(3) \ltimes \mathfrak{se}_2(3)$ with state $(\boldsymbol{X}, \boldsymbol{b}) \in \mathbb{R}^{18}$.

The third block $\boldsymbol{b}_\nu$ is **virtual**: the position row of the algebra has no sensor. It enters the flow as a velocity-like term:

$$\begin{bmatrix} R^+ \\ v^+ \\ p^+ \\ \boldsymbol{b}_g^+ \\ \boldsymbol{b}_a^+ \\ \boldsymbol{b}_\nu^+ \end{bmatrix} = \begin{bmatrix} R\exp(h[w]_\times) \\ v + hg + hR\Gamma_1(hw)a \\ p + hv + \tfrac{1}{2}h^2 g + h^2 R\Gamma_2(hw)a - hR\Gamma_1(hw)\boldsymbol{b}_\nu \\ \boldsymbol{b}_g \\ \boldsymbol{b}_a \\ \boldsymbol{b}_\nu \end{bmatrix}, \tag{27}$$

i.e. the TG filter models $\dot{p} = v - R\boldsymbol{b}_\nu$. Physically $\boldsymbol{b}_\nu \equiv 0$; the rest of the chapter is about how a filter should know that.

4.2 TG error dynamics: the constant coupling

Transport the body input by the adjoint,

$$\boldsymbol{u} = \mathrm{Ad}_{\widehat{\boldsymbol{X}}} \begin{bmatrix} w \\ a \\ -\widehat{\boldsymbol{b}_\nu} \end{bmatrix} + \begin{bmatrix} 0 \\ g \\ \hat{v} \end{bmatrix}. \tag{28}$$

The 18×18 continuous error dynamics in the TG chart are

$$\boldsymbol{A}_{\mathrm{TG}} = \begin{bmatrix} \boldsymbol{A}_{\mathrm{nav}} & -\boldsymbol{I}_9 \\ 0 & \mathrm{ad}(\boldsymbol{u}) \end{bmatrix}, \qquad \boldsymbol{A}_{\mathrm{nav}} = \begin{bmatrix} 0 & 0 & 0 \\ [g]_\times & 0 & 0 \\ 0 & \boldsymbol{I}_3 & 0 \end{bmatrix}, \qquad \boldsymbol{G}_{\mathrm{TG}} = \begin{bmatrix} -\mathrm{Ad}_{\widehat{\boldsymbol{X}}} \begin{bmatrix} \boldsymbol{I}_6 \\ 0 \end{bmatrix} & 0 \\ 0 & \mathrm{Ad}_{\widehat{\boldsymbol{X}}} \begin{bmatrix} \boldsymbol{I}_6 \\ 0 \end{bmatrix} \end{bmatrix}. \tag{29}$$

Against Eq. 24 the navigation-to-bias coupling is now $-\boldsymbol{I}_9$: no $\hat{\boldsymbol{R}}$, $\hat{v}$, $\hat{p}$. The state dependence has moved into the bias block $\mathrm{ad}(\boldsymbol{u})$ and into $\boldsymbol{G}$. The toy integrates Eq. 29 with RK4 for $\boldsymbol{F}$ and Simpson's rule for the noise columns.

4.3 TG retraction and physical Jacobian

The bias part of the TG retraction is the 9×9 version of Eq. 26:

$$\begin{bmatrix} \delta\boldsymbol{b}_g \\ \delta\boldsymbol{b}_a \\ \delta\boldsymbol{b}_\nu \end{bmatrix} = \mathrm{Ad}_{\widehat{\boldsymbol{X}}}^{-1} \; \boldsymbol{J}_{-\mathrm{ad}}(\boldsymbol{\xi}_{\mathrm{nav}}) \; \boldsymbol{\xi}_b, \qquad \boldsymbol{J}_{\mathrm{phys}}^{\mathrm{TG}} = \begin{bmatrix} \boldsymbol{J}_{\mathrm{phys}}^{\mathrm{nav}} & 0 \\ 0 & \mathrm{Ad}_{\widehat{\boldsymbol{X}}}^{-1} \end{bmatrix} \in \mathbb{R}^{18\times 18}, \tag{30}$$

with $\boldsymbol{J}_{\mathrm{phys}}^{\mathrm{nav}}$ the 9×9 block of Eq. 19. The physical chart now has 18 entries, the last three being $\delta\boldsymbol{b}_\nu$ in the body frame.

4.4 The constraint

Take the simplest case where the problem shows: identity prior navigation, corrections $\delta p = (\delta p_x, 0, 0)$ and $\delta\boldsymbol{b}_g = (0, \delta b_y, 0)$, with the native TG correction built so that its **first-order** virtual perturbation is zero. The finite retraction Eq. 30 then gives

$$\boldsymbol{b}_\nu^{\text{new}} = -\frac{1}{2}[\delta p]_\times \delta \boldsymbol{b}_g, \tag{31}$$

a non-zero virtual bias of second order. The surface $\boldsymbol{b}_\nu = 0$ is not invariant under the TG retraction: every finite correction that touches both position and gyro bias leaves it.

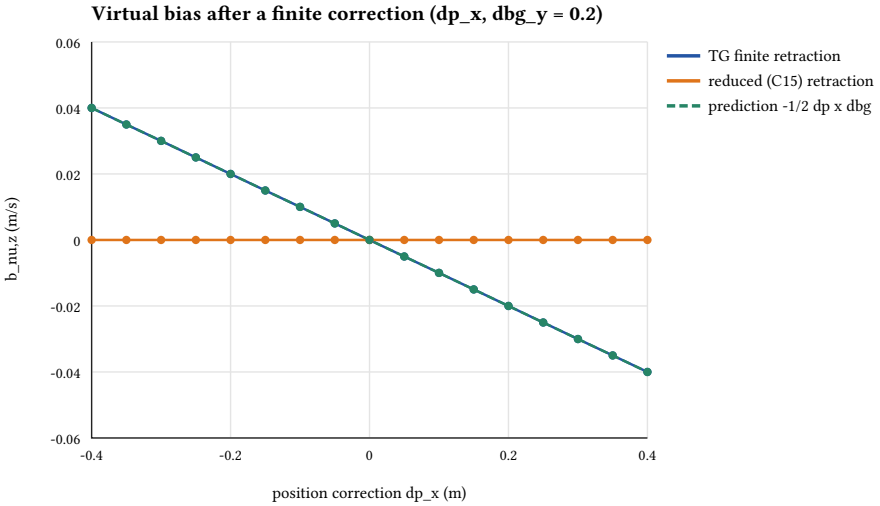


Figure 10: `constraints.csv`. The TG finite retraction follows Eq. 31 exactly (dashed prediction); the reduced C15 retraction stays at zero. This is the derived subspace only, not a global statement.

Four operations are often conflated; they are different matrices:

operation	what it does	is it a Bayesian update?
Reduced representation (C15)	state has no $\boldsymbol{b}_\nu$; the retraction never leaves the surface	yes — a different model
Finite-noise anchor	measurement $\boldsymbol{z} = 0$, $\boldsymbol{H} = [0\ 0\ 0\ 0\ 0\ I_3]$, noise $\boldsymbol{V}_a$	yes — adds evidence each time it is applied
Zero first-order tangent variance	prior $\boldsymbol{P}_{\nu\nu} = 0$; linear support restricted	only to first order; Eq. 31 reintroduces $\boldsymbol{b}_\nu$
Hard mean clamp	set $\widehat{\boldsymbol{b}_\nu} = 0$ after the update, leave $\boldsymbol{P}$	no — mean and covariance disagree

4.5 C15: the reduced embedding

C15A is the 15-dimensional filter obtained by deleting $\boldsymbol{b}_\nu$ and its row and column from the TG. Its navigation chart is Eq. 17, its biases are additive and its flow is Eq. 15 — which is Step3. The `constrainedEmbedding` test lifts a C15 correction into the TG with the compatible $\boldsymbol{\xi}_b = \boldsymbol{J}_{-\mathrm{ad}}^{-1}\,\mathrm{Ad}\big(\delta\boldsymbol{b}_g, \delta\boldsymbol{b}_a, 0\big)$ and verifies agreement to 10^{-9} with $\boldsymbol{b}_\nu = 0$: C15 is the TG restricted to the constraint surface, not an approximation of it. In `metrics.csv` Step3 and C15A agree to 10^{-8}.

4.6 Step3, Step4, Step5A, C15A in the toy

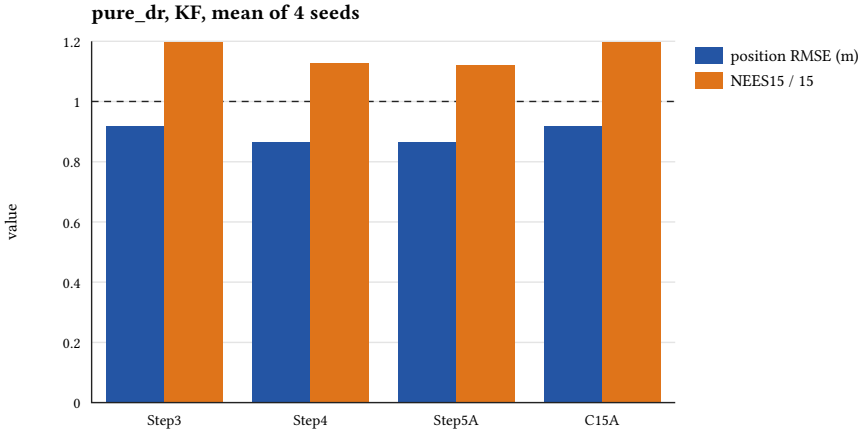


Figure 11: `pure_dr`, KF, 4-seed means. Step4 and Step5A are numerically close (0.865 vs 0.866 m; NEES 1.13 vs 1.12): the 18D TG with $\boldsymbol{b}_\nu$ free adds little over Step4 here. Step3 = C15A.

The anchor study isolates the virtual-bias treatment with a deliberately wide prior on $\boldsymbol{b}_\nu$ ($0.01\boldsymbol{I}_3$) so that it matters:

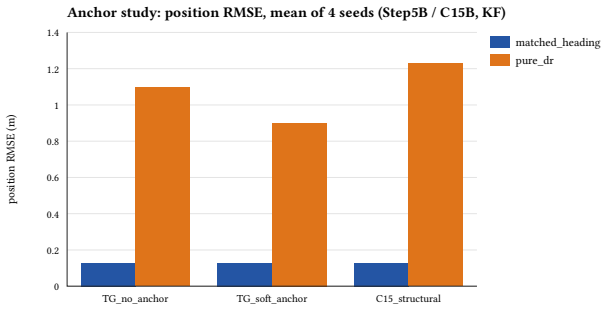


Figure 12: Position RMSE: TG without anchor, TG with a $0.001(m/s)^2$ anchor every 0.2 s, and C15. `matched_heading`: identical. `pure_dr`: 1.10 / 0.90 / 1.23 m.

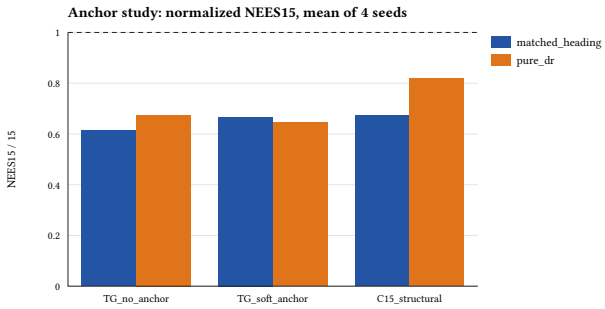


Figure 13: NEES15/15 in the same study: 0.61–0.82, all conservative because of the inflated prior.

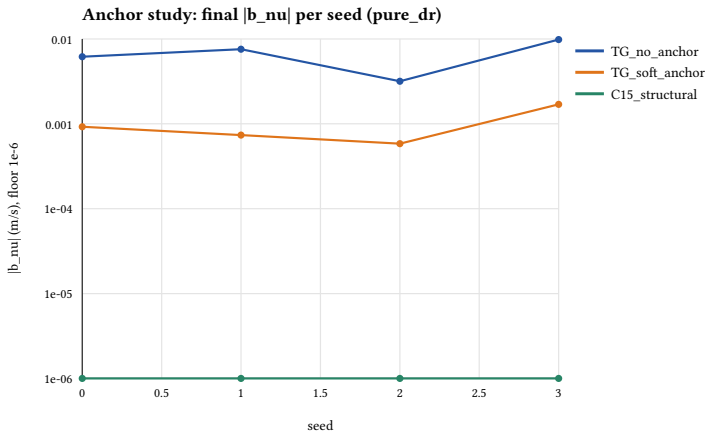


Figure 14: Final $|\boldsymbol{b}_\nu|$ per seed in `pure_dr`. Without anchor the virtual bias wanders to 3–10 $\times$ 10^{-3} m/s; the anchor holds it near 10^{-3}; C15 is structurally zero.

What the anchor study does not isolate

C15 changes the dynamics (no $\boldsymbol{b}_\nu$ term in Eq. 27), the bias retraction and the Gaussian family at once.

The `pure_dr` ordering therefore cannot be attributed to curvature of the constraint surface alone, and the archived road-data parameters are not reproduced here.

In one line
The tangent group buys a constant navigation-to-bias coupling (Eq. 29) at the price of a latent $\boldsymbol{b}_\nu$ with no sensor. A finite TG correction generically leaves $\boldsymbol{b}_\nu = 0$ (Eq. 31). The two consistent responses are to add evidence (anchor) or to remove the coordinate (C15).

5 Output approximation: the midpoint (C-star) output (Step5B, C15B)

5.1 The body-velocity measurement in the invariant chart

The velocity sensor gives $\boldsymbol{z} = R^\top v + \boldsymbol{n}$. Rotate the residual into the world frame with the current estimate and expand it in the right-invariant perturbation Eq. 17:

$$\boldsymbol{r} = \hat{\boldsymbol{R}}\boldsymbol{z} - \hat{v} = \hat{\boldsymbol{R}}R^\top v - \hat{v} = \boldsymbol{E}^\top(\boldsymbol{E}\hat{v} + \Gamma_1(\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_v) - \hat{v} = \Gamma_1(-\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_v. \tag{32}$$

To first order $\boldsymbol{r} = \boldsymbol{\xi}_v$: the Jacobian is constant. One order further,

$$\Gamma_1(-\boldsymbol{\xi}_\theta)\boldsymbol{\xi}_v = \boldsymbol{\xi}_v + \frac{1}{2}[\boldsymbol{\xi}_v]_\times\boldsymbol{\xi}_\theta + \ldots, \tag{33}$$

and replacing $\boldsymbol{\xi}_v$ by the observed $\boldsymbol{r}$ in the second-order term gives the two output models

$$\begin{bmatrix}\boldsymbol{H}_A\\ \boldsymbol{H}_B\end{bmatrix} = \begin{bmatrix}0 & \boldsymbol{I}_3 & 0 & 0 & 0\\ \frac{1}{2}[\boldsymbol{r}]_\times & \boldsymbol{I}_3 & 0 & 0 & 0\end{bmatrix}, \qquad \boldsymbol{V}_{\text{world}} = \hat{\boldsymbol{R}}\boldsymbol{V}_{\text{body}}\hat{\boldsymbol{R}}^\top. \tag{34}$$

$\boldsymbol{H}_B$ is the C-star (midpoint) output. It is the only difference between Step5A and Step5B, and between C15A and C15B, and it is applied on the single-pass KF gain row only.

5.2 Remainder orders

For a noiseless perturbed state the two residual models leave different remainders:

$$\begin{bmatrix}\boldsymbol{e}_A\\ \boldsymbol{e}_B\end{bmatrix} = \begin{bmatrix}\boldsymbol{r} - \boldsymbol{\xi}_v\\ \boldsymbol{r} - \boldsymbol{\xi}_v - \frac{1}{2}[\boldsymbol{r}]_\times\boldsymbol{\xi}_\theta\end{bmatrix} = \begin{bmatrix}O(|\boldsymbol{\xi}|^2)\\ O(|\boldsymbol{\xi}|^3)\end{bmatrix}. \tag{35}$$

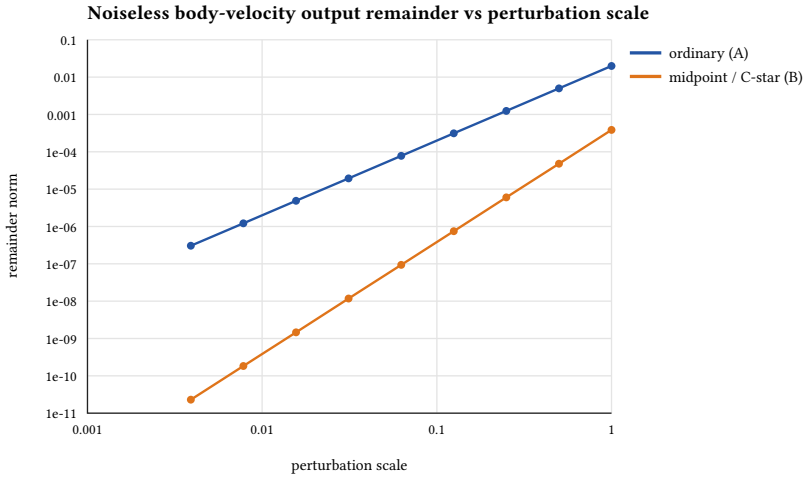


Figure 15: `remainder.csv`, direction 1. Halving the scale divides the ordinary remainder by 4 and the midpoint remainder by 8; the test checks seven directions and five scales for ratios within 4 ± 0.3 and 8 ± 0.4.

A cubic remainder is a local, noiseless statement
With noise, $\boldsymbol{H}_B$ depends on the measurement through $\boldsymbol{r}$, so the gain becomes data dependent. Nothing above implies posterior calibration or lower RMSE over a run.

5.3 A versus B in the toy (KF)

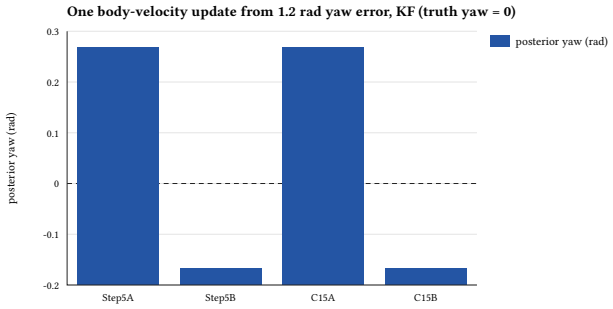


Figure 16: One body-velocity update from a 1.2 rad yaw error with a narrow velocity prior and wide yaw prior (truth 0). Output A lands at $+0.27$ rad, output B at -0.17 rad: a different single step, neither converged.

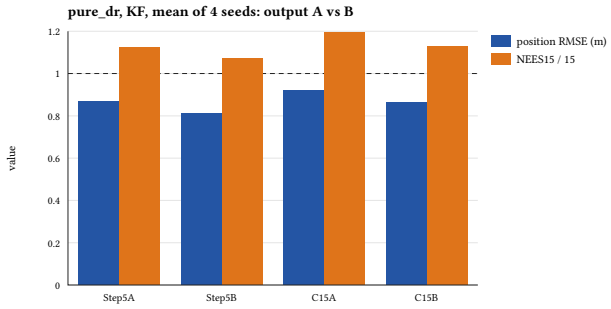


Figure 17: `pure_dr`, KF, 4-seed means. B lowers position RMSE from 0.866 to 0.813 m (Step5) and 0.919 to 0.861 m (C15) and brings NEES15/15 from 1.12/1.20 to 1.07/1.13 — within what four seeds can resolve.

In one line
C-star changes one Jacobian row (Eq. 34) and nothing else. Its effect is bounded by what a single linearization can do.

6 Iterated solver: GN and LM as fixed-prior MAP

Every recipe above was a single-pass filter. The same prior, likelihood and chart can be solved iteratively; this chapter writes the KF update as the first iterate of that solver.

6.1 The KF update is one Gauss–Newton step

Parametrize the posterior in the native tangent of the prior, whitened by the prior root $\boldsymbol{L}$ ($\boldsymbol{P}^- = \boldsymbol{L}\boldsymbol{L}^\top$):

$$\boldsymbol{x}(\boldsymbol{z}) = \mathrm{retract}(\hat{\boldsymbol{x}}^-, \boldsymbol{L}\boldsymbol{z}), \qquad C(\boldsymbol{z}) = \frac{1}{2}\boldsymbol{z}^\top\boldsymbol{z} + \frac{1}{2}\boldsymbol{r}(\boldsymbol{x}(\boldsymbol{z}))^\top \boldsymbol{V}^{-1}\boldsymbol{r}(\boldsymbol{x}(\boldsymbol{z})). \tag{36}$$

The prior enters once, as $\frac{1}{2}\boldsymbol{z}^\top\boldsymbol{z}$, and is held fixed. Linearizing the residual at the current iterate,

$$\begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{B} \\ \boldsymbol{g} \end{bmatrix} = \begin{bmatrix} \boldsymbol{H}_{\mathrm{phys}(\boldsymbol{x})} \;\; \boldsymbol{J}_{\mathrm{phys}(\boldsymbol{x})} \;\; \boldsymbol{J}_{\mathrm{reset}(\boldsymbol{L}\boldsymbol{z})} \;\; \boldsymbol{L} \\ I + \boldsymbol{A}^\top\boldsymbol{V}^{-1}\boldsymbol{A} \\ \boldsymbol{A}^\top\boldsymbol{V}^{-1}\boldsymbol{r} - \boldsymbol{z} \end{bmatrix}, \tag{37}$$

the Gauss–Newton and Levenberg–Marquardt steps are

$$\Delta\boldsymbol{z}_{\mathrm{GN}} = \boldsymbol{B}^{-1}\boldsymbol{g}, \qquad \Delta\boldsymbol{z}_{\mathrm{LM}} = (\boldsymbol{B} + \lambda\,\mathrm{diag}(\boldsymbol{B}))^{-1}\boldsymbol{g}, \tag{38}$$

with λ multiplied by 10 after a rejected trial and divided by 3 after an accepted one. At $\boldsymbol{z} = 0$ one GN step equals the Kalman correction of Eq. 8 (matrix-inversion lemma on $\boldsymbol{B}^{-1}$): the KF is the first iterate.

6.2 Posterior covariance

After the last accepted iterate the covariance is formed from the undamped information and mapped once through the final reset:

$$\boldsymbol{P}^+ = \boldsymbol{J}_{\mathrm{reset}(\boldsymbol{L}\boldsymbol{z})}\;\boldsymbol{L}\;\boldsymbol{B}^{-1}\;\boldsymbol{L}^\top\;\boldsymbol{J}_{\mathrm{reset}(\boldsymbol{L}\boldsymbol{z})}^\top. \tag{39}$$

LM damping stabilizes the search but never shrinks the posterior. Stopping statuses — `gradient`, `step`, `cost`, `stalled`, `capped`, `time_budget` — are recorded per update; only the first three mean convergence.

6.3 Why the prior must stay fixed

Replacing the prior by the previous iterate's posterior would add $\frac{1}{2}\boldsymbol{z}^\top\boldsymbol{z}$ again with a smaller $\boldsymbol{L}$: the same measurement would be counted at every iteration and the covariance would collapse. The `gaussian` test enforces the linear case: KF, GN and LM must all return $\boldsymbol{P}^+ = \left(\boldsymbol{P}^{-1} + \boldsymbol{H}^\top\boldsymbol{V}^{-1}\boldsymbol{H}\right)^{-1}$ regardless of iteration count.

6.4 GN/LM across all recipes in the toy

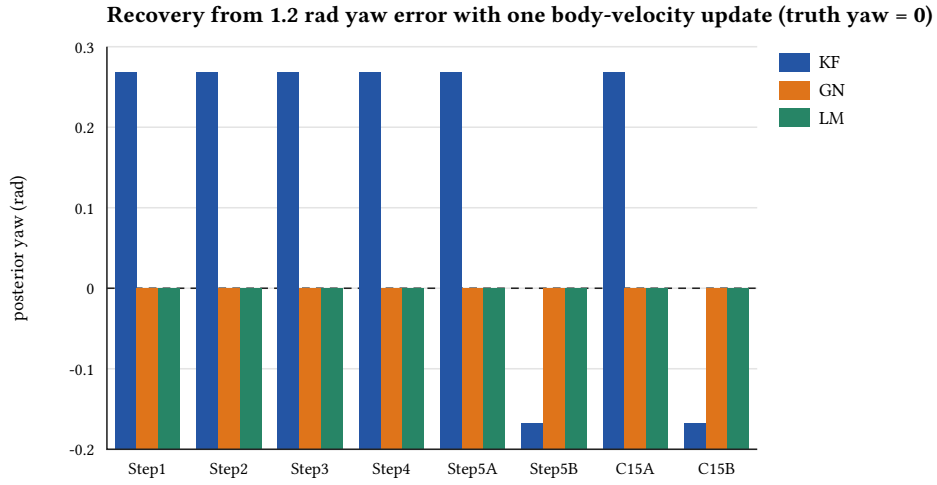


Figure 18: `recovery.csv`: the recovery problem of Section 5 for all recipes and backends. GN and LM take 4–5 iterations from cost 1020 to 0.36 and reach 4×10^{-4} rad for every recipe; the chart and the output variant stop mattering.

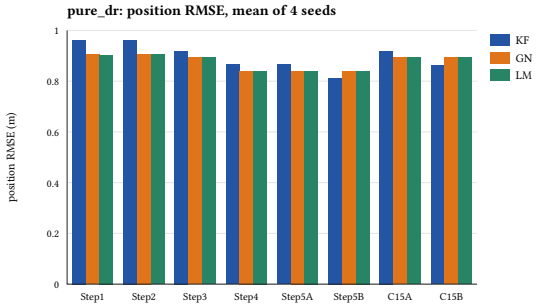


Figure 19: `pure_dr`, 4-seed means, all recipes and backends. GN/LM lower RMSE by 3–6% and make A and B identical (rows agree to 10^{-8}).

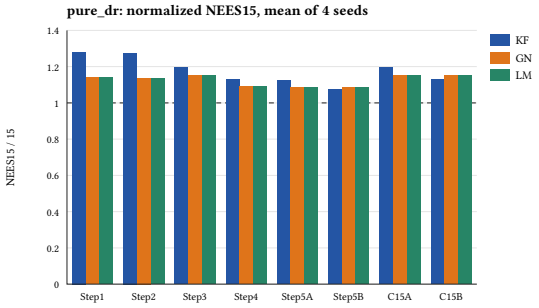


Figure 20: `pure_dr` NEES15/15. Iteration moves every recipe 0.05–0.15 toward 1; none reach it. In `matched_heading` GN/LM change nothing measurable.

In one line
GN/LM = repeat the Kalman step with the same prior and an updated linearization (Eq. 37–Eq. 38), then report Eq. 39. They solve the stated problem better; they do not change what it assumes.

7 Reading the diagnostics: when the model is wrong

7.1 Cost is not truth

One heading update from prior yaw 1.2 rad, truth 0, measurement noise 0.05 rad:

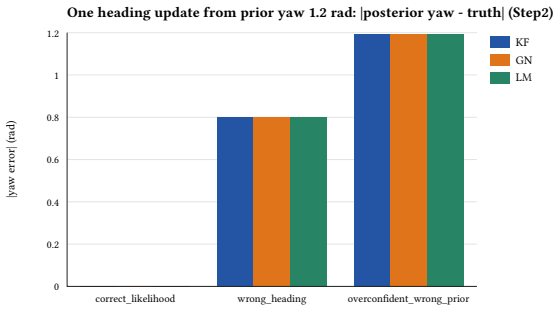


Figure 21: Posterior yaw error. Correct likelihood: 0.0015 rad. Heading biased by 0.8 rad: the filter lands at 0.80, where the data says. Overconfident prior (10^{-5}rad^2): the correct measurement is ignored, 1.195 rad remains.

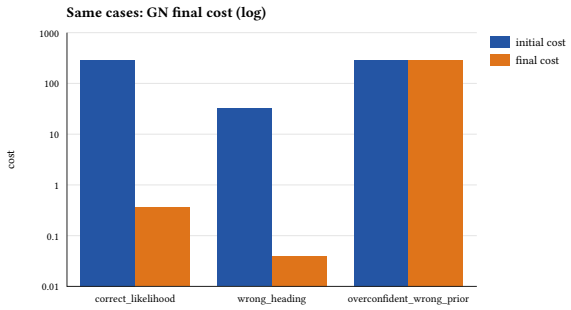


Figure 22: GN cost before and after for the same cases. The wrong-heading case has the **lowest** final cost (0.04): the optimizer solved its problem perfectly. The overconfident case barely moves (288 ⟶ 287).

Identical numbers appear for every recipe and backend: no chart or solver can hide a wrong assumption.

7.2 An unmodelled bias step

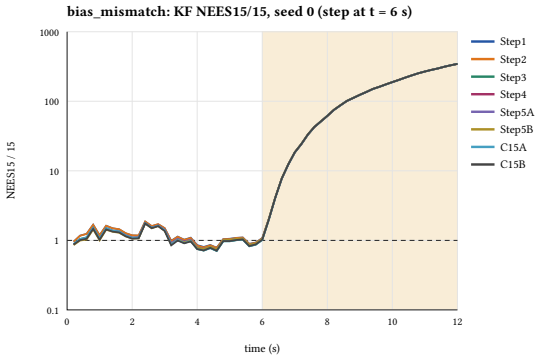


Figure 23: NEES15/15 after a +0.12 rad/s gyro-z step at 6 s with zero bias random walk in the filter. All recipes rise to ≈300 by 12 s: the covariance has frozen while the error grows.

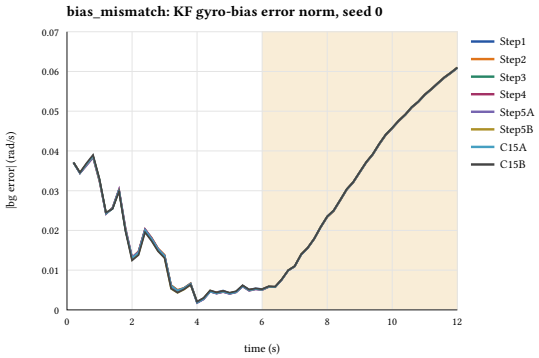


Figure 24: Gyro-bias error norm: the filter follows the step slowly (0.05 rad/s remaining after 6 s) because its process model says the bias cannot move.

`metrics.csv` reports NEES15/15 ≈ 72 for every recipe and backend. Invariant charts, tangent groups and iterated solvers all give the same answer because none of them touch the process-noise model.

7.3 Full 3-D velocity versus forward + NHC

What the non-holonomic constraint is, and why it is assumed
A ground vehicle's speed sensor (wheel odometry, CAN) measures one number: the forward speed along the body x-axis. The velocity row of Eq. 10 wants three. The **non-holonomic constraint** (NHC) fills the missing two with a kinematic assumption about wheeled motion: a vehicle that does not skid sideways and stays on the road surface has zero lateral and zero vertical velocity **at the wheel contact point, in the body frame**. The constraint is written as a pseudo-measurement that reuses Eq. 10 unchanged,

$$\boldsymbol{z}_{\text{NHC}} = \begin{bmatrix} \tilde{v}_x \\ 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{y} = \hat{\boldsymbol{R}}^{\top} \hat{v}, \qquad \boldsymbol{V}_{\text{NHC}} = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_{\text{lat}}^2 & 0 \\ 0 & 0 & \sigma_{\text{vert}}^2 \end{bmatrix}, \tag{40}$$

so the filter sees a full 3-D velocity although only one channel was sensed. The assumption is made because it is cheap and powerful: the two zero channels are what ties the IMU to the vehicle. Through the $[\hat{v}_B]_\times$ block of Eq. 10 the lateral channel observes yaw-rate-related attitude and gyro-bias error, and the vertical channel observes pitch and the vertical velocity — exactly the directions that dead reckoning on forward speed alone leaves unobserved. The price is that $\sigma_{\text{lat}}, \sigma_{\text{vert}}$ are not sensor specifications but a statement about the vehicle: they must cover sideslip in fast cornering or low friction, vertical motion on ramps and bumps, and the lever arm between the IMU and the axle (the IMU's own lateral velocity in a turn is $\omega_z l$, not zero). In this note the toy sets all three variances to $0.0025(m/s)^2$ and asks what happens when the vehicle does and does not obey the assumption.

Step2, truth with lateral body velocity 0 or 0.6 m/s, heading always available, each velocity channel variance $0.0025(m/s)^2$. The NHC variant Eq. 40 uses the same noisy forward sample and sets the lateral and vertical channels to zero.

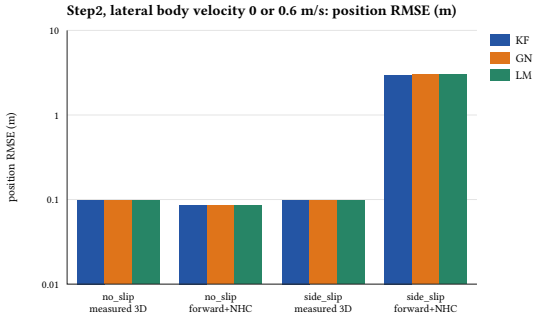


Figure 25: Log scale. Without sideslip NHC is a valid constraint and slightly helps (0.086 vs 0.099 m). With 0.6 m/s sideslip the NHC likelihood is wrong: 2.98 m.

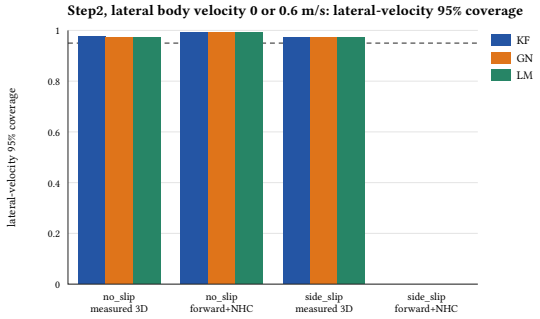


Figure 26: Lateral-velocity 95% coverage. Measured 3-D: 0.97. NHC without slip: 0.99. NHC with slip: 0.00 — confidently wrong in the lateral direction.

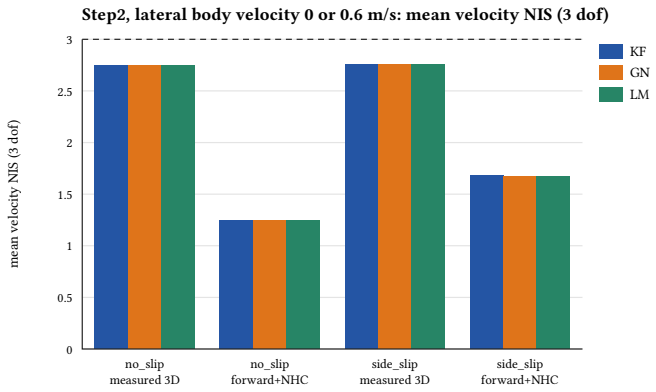


Figure 27: Mean velocity NIS (3 d.o.f.). NHC **lowers** NIS (1.2–1.7 vs 2.75) even when wrong, because two of its three channels are noiseless by construction; NIS alone would not flag the failure. Coverage against truth does.

In one line

NEES and coverage against truth expose model mismatch; cost and NIS do not. The remedies — bias random walk, sideslip in the measurement model — are in the model, not in the geometry.

Appendix A: Common ground: state, operators and the yardstick

A.1 State, frames and IMU model

A world frame W and a body (IMU) frame B; R maps body to world. Gravity is a fixed parameter.

$$\boldsymbol{x} = \big(R, v, p, \boldsymbol{b}_g, \boldsymbol{b}_a\big), \qquad R \in \mathrm{SO}(3), \quad v, p \in \mathbb{R}^3 \text{ (world)}, \quad \boldsymbol{b}_g, \boldsymbol{b}_a \in \mathbb{R}^3 \text{ (body)}, \tag{41}$$

$$g = \begin{bmatrix} 0 \\ 0 \\ -9.81 \end{bmatrix} \text{ (fixed parameter)}. \tag{42}$$

IMU samples are gyro $\tilde{w}$ and specific force $\tilde{a}$ in the body frame, with white noise $\boldsymbol{n}$ and random-walk biases:

$$\begin{bmatrix} \tilde{w} \\ \tilde{a} \end{bmatrix} = \begin{bmatrix} w \\ a \end{bmatrix} + \begin{bmatrix} \boldsymbol{b}_g \\ \boldsymbol{b}_a \end{bmatrix} + \begin{bmatrix} \boldsymbol{n}_g \\ \boldsymbol{n}_a \end{bmatrix}, \qquad \begin{bmatrix} \dot{\boldsymbol{b}}_g \\ \dot{\boldsymbol{b}}_a \end{bmatrix} = \begin{bmatrix} \boldsymbol{n}_{bg} \\ \boldsymbol{n}_{ba} \end{bmatrix}. \tag{43}$$

Time is t; one propagation interval has length h. A hat marks an estimate; the same symbol without a hat is the true value; a superscript $+$ is the value after a propagation or update.

A.2 Operators

Skew matrix. For $\boldsymbol{x} \in \mathbb{R}^3$,

$$[\boldsymbol{x}]_\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \qquad [\boldsymbol{x}]_\times \boldsymbol{y} = \boldsymbol{x} \times \boldsymbol{y}. \tag{44}$$

Rotation exponential and the Γ matrices. With $\boldsymbol{\varphi} \in \mathbb{R}^3$ and $\boldsymbol{\Phi} = [\boldsymbol{\varphi}]_\times$,

$$\exp(\boldsymbol{\Phi}) = \sum_{k \ge 0} \frac{\boldsymbol{\Phi}^k}{k!}, \qquad \Gamma_1(\boldsymbol{\varphi}) = \sum_{k \ge 0} \frac{\boldsymbol{\Phi}^k}{(k+1)!}, \qquad \Gamma_2(\boldsymbol{\varphi}) = \sum_{k \ge 0} \frac{\boldsymbol{\Phi}^k}{(k+2)!}. \tag{45}$$

$\exp(\boldsymbol{\Phi})$ is a rotation matrix; Γ_1 is the left Jacobian of SO(3), so $\Gamma_1(-\boldsymbol{\varphi})$ is the right Jacobian $\boldsymbol{J}_r(\boldsymbol{\varphi})$; Γ_2 is its second integral. The toy evaluates all three by the series (`gammas`) and the tests compare them with closed forms.

Adjoint and little adjoint of the extended pose. For a navigation triple (R, v, p) and a tangent vector $\boldsymbol{\xi} = (\boldsymbol{\xi}_\theta, \boldsymbol{\xi}_v, \boldsymbol{\xi}_p)$,

$$\mathrm{Ad}_{R,v,p} = \begin{bmatrix} R & 0 & 0 \\ [v]_\times R & R & 0 \\ [p]_\times R & 0 & R \end{bmatrix}, \qquad \mathrm{ad}(\boldsymbol{\xi}) = \begin{bmatrix} [\boldsymbol{\xi}_\theta]_\times & 0 & 0 \\ [\boldsymbol{\xi}_v]_\times & [\boldsymbol{\xi}_\theta]_\times & 0 \\ [\boldsymbol{\xi}_p]_\times & 0 & [\boldsymbol{\xi}_\theta]_\times \end{bmatrix}. \tag{46}$$

The 6×6 restriction to (R, v) drops the last block row and column.

A.3 The physical chart and the scores

Each recipe stores its covariance in its own coordinates. To compare them, every covariance is mapped once into a physical error vector of fixed meaning,

$$e = \begin{bmatrix} \mathrm{Log}\left(\hat{\boldsymbol{R}}^\top R\right) \\ v - \hat{v} \\ p - \hat{p} \\ \boldsymbol{b}_g - \hat{\boldsymbol{b}_g} \\ \boldsymbol{b}_a - \widehat{\boldsymbol{b}_a} \end{bmatrix} \in \mathbb{R}^{15}, \qquad \text{ordered as } \left(\delta\boldsymbol{\theta}_{\text{body}}, \delta v_{\text{world}}, \delta p_{\text{world}}, \delta\boldsymbol{b}_g, \delta\boldsymbol{b}_a\right), \tag{47}$$

and all scores are computed there:

$$\begin{bmatrix} \mathrm{NEES}_{15} \\ \mathrm{coverage}_z \\ \mathrm{NIS} \end{bmatrix} = \begin{bmatrix} \boldsymbol{e}^\top \boldsymbol{P}_{\text{phys}}^{-1} \boldsymbol{e} \\ \Pr\left(|e_z| \le 1.96\sqrt{P_{zz}}\right) \\ \boldsymbol{r}^\top (\boldsymbol{H}\boldsymbol{P}\boldsymbol{H}^\top + \boldsymbol{V})^{-1} \boldsymbol{r} \end{bmatrix}. \tag{48}$$

`normalized_nees15` divides the time average of NEES_{15} by 15; a consistent filter sits near 1. NIS is read after the velocity update (3 d.o.f., expect ≈ 3). Pose RMSE is direct world-frame error without alignment.

A.4 The common experiment

All scenarios use one closed-form, variable-speed circle; truth never calls the estimator.

$$\begin{bmatrix} R(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} R_z(0.3t) \\ R(t) \begin{bmatrix} 2 + 0.3\sin 0.7t \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}, \tag{49}$$

with IMU at 40 Hz (interval midpoint sampled and held) and body velocity and heading every 0.2 s. Noise is seeded `mt19937` with an explicit Box–Muller transform; all recipes receive the same samples and the same prior.

scenario	assumption	what to look at
`matched_heading`	fixed biases; matched IMU / velocity / heading noise	baseline errors, NEES, coverage
`heading_dropout`	heading absent from 35% to 75% of the run	attitude/bias uncertainty growth and recovery
`pure_dr`	IMU and full 3-D body velocity only	global yaw is unobservable
`bias_mismatch`	+0.12 rad/s gyro-z step at mid-run, zero bias random walk in the filter	a converged filter can be confidently wrong

Prior errors are drawn in the physical chart with per-axis standard deviations

$$\begin{bmatrix} \sigma_\theta \\ \sigma_v \\ \sigma_p \\ \sigma_{bg} \\ \sigma_{ba} \end{bmatrix} = \begin{bmatrix} 0.05\ \text{rad} \\ 0.1\ \text{m/s} \\ 0.1\ \text{m} \\ 0.02\ \text{rad/s} \\ 0.05\text{m/s}^2 \end{bmatrix}. \tag{50}$$

Four seeds are illustrative, not statistics; time samples within a seed are correlated.

Appendix B: Symbol to code

symbol / equation	function	file
$[\cdot]_\times$, exp, Log, Γ_1, Γ_2 (Eq. 44, Eq. 45)	`hat` , `expSO3` , `logSO3` , `gammas`	`src/kf3d/geometry.cpp`
Ad, ad, $\sum \mathrm{ad}^k/(k+1)!$ (Eq. 46)	`adjoint` , `ad` , `algebraJacobian`	`src/kf3d/geometry.cpp`

charts Eq. 2, Eq. 18, Eq. 26, Eq. 30	`retract`, `local`	`src/kf3d/geometry.cpp`
$\boldsymbol{J}_{\text{phys}}$ Eq. 19, $\boldsymbol{J}_{\text{reset}}$ Eq. 20	`physicalJacobian`, `resetJacobian`	`src/kf3d/geometry.cpp`
flows Eq. 1, Eq. 15; $\boldsymbol{F}$ Eq. 3, Eq. 16	`physicalFlow`	`src/kf3d/propagation.cpp`
$\boldsymbol{G}$ Eq. 4, $\boldsymbol{Q}$ Eq. 5, conjugation Eq. 21	`noiseRoot`, `predict`	`src/kf3d/propagation.cpp`
TG flow Eq. 27, $\boldsymbol{A}_{\text{TG}}$ Eq. 29, RK4	`tgFlow`, `dynamics`, `transition`	`src/kf3d/propagation.cpp`
Joseph Eq. 8, reset Eq. 9, QR root Eq. 7	`update`, `compress`	`src/kf3d/filter.cpp`
$\boldsymbol{H}_A$, $\boldsymbol{H}_B$ Eq. 34	`observeVelocity`	`src/kf3d/filter.cpp`
cost Eq. 36, GN/LM Eq. 37–Eq. 38, posterior Eq. 39	`update` (non-KF branch), `GnSolver`, `LmSolver`	`src/kf3d/filter.cpp`
anchor $\boldsymbol{H} = [0 \ \dots \ I_3]$	`observeVirtualBias`	`src/kf3d/filter.cpp`
scenario and study generators	`events`, `*Csv`	`examples/teaching.cpp`
analytic / finite-difference oracles	`geometry`, `zoh`, `gaussian`, `prediction`, `equivalence`, `remainder`	`tests/math_contracts.cpp`

Appendix C: Regenerating the figures

```
./build/kf_math_tests                          # analytic + finite-difference contracts
./build/kf_teaching results/teaching 4 12      # all CSVs used in this note
python3 docs/handnote/make_figs.py               # fig/*.svg, metrics_mean.csv
typst compile docs/handnote/handnote.typ         # this PDF (Typst 0.15)
```

The figures in this note were drawn from the CSVs with a standard-library SVG script (`make_figs.py`, no third-party packages); no estimator code was changed. The same binary, platform and arguments reproduce the CSV bytes; the four-seed means quoted in captions are rounded from those files.

Appendix D: Mean metrics, 4 seeds, 12 s

scenario	recipe	backend	pos RMSE (m)	att RMSE (rad)	NEES15/15	vel NIS
matched_heading	Step1	KF	0.1792	0.01620	0.954	2.901
matched_heading	Step1	GN	0.1852	0.01625	0.954	2.898
matched_heading	Step1	LM	0.1852	0.01625	0.954	2.898
matched_heading	Step2	KF	0.1808	0.01618	0.953	2.902
matched_heading	Step2	GN	0.1868	0.01623	0.953	2.898
matched_heading	Step2	LM	0.1868	0.01623	0.953	2.898
matched_heading	Step3	KF	0.1851	0.01620	0.944	2.895
matched_heading	Step3	GN	0.1876	0.01621	0.945	2.893
matched_heading	Step3	LM	0.1876	0.01621	0.945	2.893
matched_heading	Step4	KF	0.1862	0.01617	0.944	2.896
matched_heading	Step4	GN	0.1886	0.01618	0.944	2.895
matched_heading	Step4	LM	0.1886	0.01618	0.944	2.895
matched_heading	Step5A	KF	0.1893	0.01617	0.946	2.896
matched_heading	Step5A	GN	0.1917	0.01618	0.946	2.895
matched_heading	Step5A	LM	0.1917	0.01618	0.946	2.895
matched_heading	Step5B	KF	0.1917	0.01618	0.945	2.892
matched_heading	Step5B	GN	0.1917	0.01618	0.946	2.895
matched_heading	Step5B	LM	0.1917	0.01618	0.946	2.895
matched_heading	C15A	KF	0.1851	0.01620	0.944	2.895
matched_heading	C15A	GN	0.1876	0.01621	0.945	2.893
matched_heading	C15A	LM	0.1876	0.01621	0.945	2.893
matched_heading	C15B	KF	0.1876	0.01621	0.944	2.891
matched_heading	C15B	GN	0.1876	0.01621	0.945	2.893

matched_heading	C15B	LM	0.1876	0.01621	0.945	2.893
heading_dropout	Step1	KF	0.1945	0.02266	0.959	2.901
heading_dropout	Step1	GN	0.1999	0.02259	0.959	2.897
heading_dropout	Step1	LM	0.1999	0.02259	0.959	2.897
heading_dropout	Step2	KF	0.1960	0.02266	0.958	2.901
heading_dropout	Step2	GN	0.2015	0.02259	0.957	2.897
heading_dropout	Step2	LM	0.2015	0.02259	0.957	2.897
heading_dropout	Step3	KF	0.1998	0.02263	0.948	2.894
heading_dropout	Step3	GN	0.2021	0.02258	0.948	2.892
heading_dropout	Step3	LM	0.2021	0.02258	0.948	2.892
heading_dropout	Step4	KF	0.2013	0.02272	0.948	2.895
heading_dropout	Step4	GN	0.2035	0.02267	0.948	2.893
heading_dropout	Step4	LM	0.2035	0.02267	0.948	2.893
heading_dropout	Step5A	KF	0.2047	0.02272	0.950	2.895
heading_dropout	Step5A	GN	0.2069	0.02267	0.950	2.893
heading_dropout	Step5A	LM	0.2069	0.02267	0.950	2.893
heading_dropout	Step5B	KF	0.2068	0.02263	0.949	2.891
heading_dropout	Step5B	GN	0.2069	0.02267	0.950	2.893
heading_dropout	Step5B	LM	0.2069	0.02267	0.950	2.893
heading_dropout	C15A	KF	0.1998	0.02263	0.948	2.894
heading_dropout	C15A	GN	0.2021	0.02258	0.948	2.892
heading_dropout	C15A	LM	0.2021	0.02258	0.948	2.892
heading_dropout	C15B	KF	0.2020	0.02254	0.947	2.890
heading_dropout	C15B	GN	0.2021	0.02258	0.948	2.892
heading_dropout	C15B	LM	0.2021	0.02258	0.948	2.892
pure_dr	Step1	KF	0.9621	0.11476	1.281	2.915
pure_dr	Step1	GN	0.9053	0.10878	1.141	2.913
pure_dr	Step1	LM	0.9031	0.10853	1.139	2.913
pure_dr	Step2	KF	0.9627	0.11482	1.276	2.915
pure_dr	Step2	GN	0.9074	0.10898	1.137	2.913
pure_dr	Step2	LM	0.9049	0.10872	1.135	2.913
pure_dr	Step3	KF	0.9188	0.10925	1.196	2.908
pure_dr	Step3	GN	0.8948	0.10713	1.152	2.906
pure_dr	Step3	LM	0.8948	0.10713	1.152	2.906
pure_dr	Step4	KF	0.8652	0.10239	1.128	2.911
pure_dr	Step4	GN	0.8397	0.09989	1.092	2.909
pure_dr	Step4	LM	0.8397	0.09989	1.092	2.909
pure_dr	Step5A	KF	0.8659	0.10239	1.122	2.911
pure_dr	Step5A	GN	0.8402	0.09989	1.086	2.909
pure_dr	Step5A	LM	0.8402	0.09989	1.086	2.909
pure_dr	Step5B	KF	0.8126	0.09805	1.072	2.906
pure_dr	Step5B	GN	0.8402	0.09989	1.086	2.909
pure_dr	Step5B	LM	0.8402	0.09989	1.086	2.909
pure_dr	C15A	KF	0.9188	0.10925	1.196	2.908
pure_dr	C15A	GN	0.8948	0.10713	1.152	2.906
pure_dr	C15A	LM	0.8948	0.10713	1.152	2.906
pure_dr	C15B	KF	0.8612	0.10459	1.128	2.904
pure_dr	C15B	GN	0.8948	0.10713	1.152	2.906
pure_dr	C15B	LM	0.8948	0.10713	1.152	2.906
bias_mismatch	Step1	KF	0.3172	0.09102	72.610	3.078
bias_mismatch	Step1	GN	0.3160	0.09098	72.605	3.076
bias_mismatch	Step1	LM	0.3160	0.09098	72.605	3.076
bias_mismatch	Step2	KF	0.3180	0.09102	72.596	3.075
bias_mismatch	Step2	GN	0.3169	0.09098	72.591	3.073
bias_mismatch	Step2	LM	0.3169	0.09098	72.591	3.073
bias_mismatch	Step3	KF	0.3576	0.09087	72.774	3.068
bias_mismatch	Step3	GN	0.3565	0.09085	72.774	3.068
bias_mismatch	Step3	LM	0.3565	0.09085	72.774	3.068

bias_mismatch	Step4	KF	0.3569	0.09083	72.806	3.067
bias_mismatch	Step4	GN	0.3557	0.09081	72.804	3.066
bias_mismatch	Step4	LM	0.3557	0.09081	72.804	3.066
bias_mismatch	Step5A	KF	0.3845	0.09083	72.771	3.067
bias_mismatch	Step5A	GN	0.3831	0.09081	72.768	3.066
bias_mismatch	Step5A	LM	0.3831	0.09081	72.768	3.066
bias_mismatch	Step5B	KF	0.3788	0.09083	72.743	3.064
bias_mismatch	Step5B	GN	0.3831	0.09081	72.768	3.066
bias_mismatch	Step5B	LM	0.3831	0.09081	72.768	3.066
bias_mismatch	C15A	KF	0.3576	0.09087	72.774	3.068
bias_mismatch	C15A	GN	0.3565	0.09085	72.774	3.068
bias_mismatch	C15A	LM	0.3565	0.09085	72.774	3.068
bias_mismatch	C15B	KF	0.3523	0.09088	72.749	3.066
bias_mismatch	C15B	GN	0.3565	0.09085	72.774	3.068
bias_mismatch	C15B	LM	0.3565	0.09085	72.774	3.068